



NORTH SYDNEY BOYS HIGH SCHOOL

2011 HSC ASSESSMENT TASK 2

Mathematics

General Instructions

- Working time – 55 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Ms Ziariaris
- ☐ Mr Law
- ☐ Mr Weiss
- ☐ Mr Lam
- ☐ Mr Ireland
- ☐ Mr Fletcher
- ☐ Ms Collins/Mr Rezcallah

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total %
Mark	22	18	12	4	56	100

Question 1 (22 marks)

a) Find.

i. $\int \frac{7}{x^2} dx$ 2

ii. $\int \frac{5x^2 + x^3}{x^2} dx$ 2

iii. $\int \sqrt{x-1} dx$ 2

b) Evaluate.

i. $\int_{-8}^8 x^3 dx$ 1

ii. $\int_2^6 (x+1)(5x-2) dx$ 2

iii. $\int_1^4 x^2 \sqrt{x} dx$ 2

c)

i. Sketch the graph of $y = 16 - x^2$. 2

ii. Approximate the area under this graph between $x=1$ and $x=3$ and the x axis using the trapezoidal rule with three function values. 3

iii. Integrate the same section of the graph and find an exact value. 2

iv. Explain why the areas to part ii and iii are different. 1

d) Find the value of k .

$\int_2^k (x+3) dx = 30$ 3

Question 2 (18 marks) Start A New Page

a) The quadratic $2x^2 + 5x + 1 = 0$ has roots α and β .

Without finding the roots, evaluate:

- | | |
|--|---|
| i. $\alpha + \beta$ | 1 |
| ii. $\alpha \beta$ | 1 |
| iii. $\alpha^2 + \beta^2$ | 2 |
| iv. $\frac{1}{\alpha} + \frac{1}{\beta}$ | 1 |

b) Given the roots of a quadratic equation are: $\alpha = 3 + \sqrt{5}$ and $\beta = 3 - \sqrt{5}$. Write the equation in the form $ax^2 + bx + c = 0$. 3

c) Solve: 3

$$x^4 - 17x^2 + 16 = 0$$

d) Find the values of A, B and C 3

$$5x^2 + 6x - 7 \equiv A(x - 2)^2 + B(x - 2) + C$$

e) Given the quadratic function

$$y = 2x^2 + kx + 8$$

- | | |
|---|---|
| i. For what values of k does the function pass through the point (2,0)? | 1 |
| ii. For what values of k is the equation positive definite. | 2 |
| iii. Can the function be negative definite? Why? | 1 |

Question 3 (12 marks) Start A New Page

a) Given the parabola.

$$y = (x - 1)^2 - 3$$

Find:

- | | |
|--|---|
| i. the vertex | 1 |
| ii. the focus | 2 |
| iii. the equation of the directrix | 1 |
| iv. Sketch the parabola, indicating clearly the directrix, vertex and focus. | 2 |

b) Find the equation of the locus of a point, P, that is equidistant from two points A(4, 6) and B(8, 10) 2

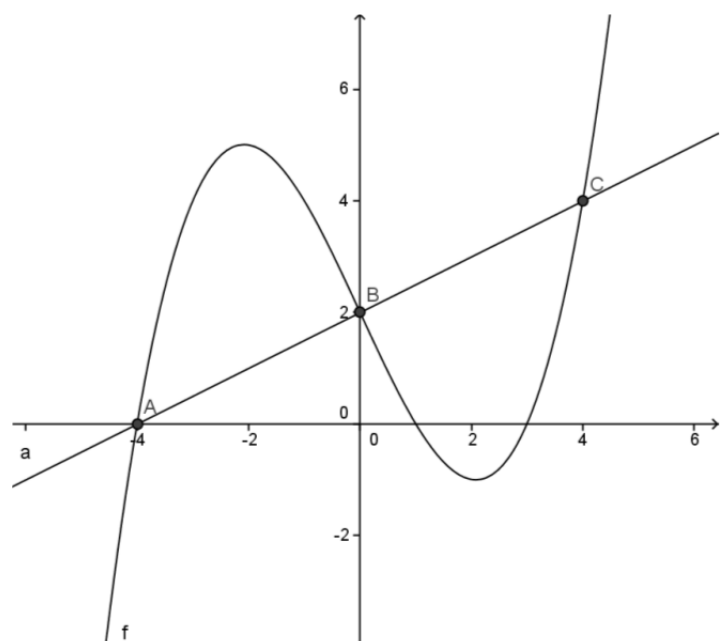
- c)
- | | |
|---|---|
| i) Find the locus of a point P(x,y) which moves so its distance from point A (-3,3) is twice the distance from the point B (2, -9). | 2 |
| ii) Describe this locus geometrically. | 2 |

Question 4 (4 marks) Start A New Page

The curves $y = \frac{1}{6}(x - 3)(x - 1)(x + 4)$ and $y = 2x + 2$ intersect at 4

the points A(-4,0), B(0,2) and C(4,4) as shown in the diagram.

Calculate the area enclosed between the two curves.



Suggested Solutions

Question 1

(a) i. (2 marks)

$$\begin{aligned}\int \frac{7}{x^2} dx &= \int 7x^{-2} dx \\ &= -7x^{-1} + C\end{aligned}$$

ii. (2 marks)

$$\begin{aligned}\int \frac{5x^2 + x^3}{x^2} dx &= \int \left(\frac{5\cancel{x^2}}{\cancel{x^2}} + \frac{x^{\cancel{3}}}{\cancel{x^2}} \right) dx \\ &= \int (5 + x) dx \\ &= 5x + \frac{1}{2}x^2 + C\end{aligned}$$

iii. (2 marks)

$$\begin{aligned}\int \sqrt{x-1} dx &= \int (x-1)^{\frac{1}{2}} dx \\ &= \frac{2}{3}(x-1)^{\frac{3}{2}} + C\end{aligned}$$

(b) i. (1 mark)

$$\int_{-8}^8 x^3 dx = 0$$

(integral of odd function over balanced interval centred around the origin)

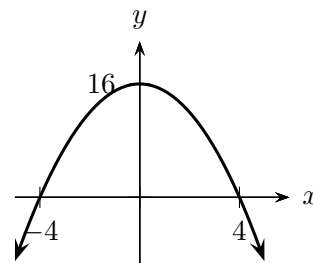
ii. (2 marks)

$$\begin{aligned}&\int_2^6 (x+1)(5x-2) dx \\ &= \int_2^6 (5x^2 + 3x - 2) dx \\ &= \left[\frac{5x^3}{3} + \frac{3x^2}{2} - 2x \right]_2^6 \\ &= \frac{5}{3}(6^3 - 2^3) + \frac{3}{2}(6^2 - 2^2) \\ &\quad - 2(6 - 2) \\ &= \frac{1160}{3}\end{aligned}$$

iii. (2 marks)

$$\begin{aligned}\int_1^4 x^2 \sqrt{x} dx &= \int_1^4 x^2 \times x^{\frac{1}{2}} dx \\ &= \int_1^4 x^{\frac{5}{2}} dx \\ &= \left[\frac{2}{7} x^{\frac{7}{2}} \right]_1^4 \\ &= \frac{2}{7} (4^{\frac{7}{2}} - 1) \\ &= \frac{254}{7}\end{aligned}$$

(c) i. (2 marks)



ii. (3 marks)

$$\begin{array}{c|c|c|c} x & 1 & 2 & 3 \\ \hline 16-x^2 & 15 & 12 & 7 \end{array}$$

$$\begin{aligned}A &\approx \frac{h}{2} \left(y_1 + 2 \sum y_{\text{middle}} + y_\ell \right) \\ &= \frac{1}{2} (15 + 2(12) + 7) \\ &= 23\end{aligned}$$

iii. (2 marks)

$$\begin{aligned}&\int_1^3 16 - x^2 dx \\ &= \left[16x - \frac{1}{3}x^3 \right]_1^3 \\ &= 16(3-1) - \frac{1}{3}(3^3-1) \\ &= \frac{70}{3} \approx 23.33\end{aligned}$$

iv. (1 mark)

The areas are different as the trapezoidal rule only gives an approximation to the exact area.

(d) (3 marks)

$$\begin{aligned}\int_2^k (x+3) dx &= \left[\frac{1}{2}x^2 + 3x \right]_2^k \\ &= \frac{1}{2}(k^2 - 2^2) + 3(k-2) \\ &= 30\end{aligned}$$

$$\therefore k^2 - 4 + 6(k-2) = 60$$

$$k^2 + 6k - 16 = 60$$

$$k^2 + 6k + 9 = 76 + 9 = 85$$

$$(k+3)^2 = 85$$

$$k+3 = \pm\sqrt{85}$$

$$\therefore k = -3 \pm \sqrt{85}$$

Let $m = x^2$,

$$m^2 - 17m + 16 = 0$$

$$(m-16)(m-1) = 0$$

$$m = 1, 16 \therefore x^2 = 1, 16$$

$$\therefore x = \pm 1, \pm 4$$

(d) (3 marks)

$$5x^2 + 6x - 7 \equiv A(x-2)^2 + B(x-2) + C$$

By inspection, $A = 5$. Letting $x = 2$,

$$5(2^2) + 6(2) - 7 = C$$

$$C = 20 + 12 - 7 = 25$$

Question 2(a) $2x^2 + 5x + 1 = 0$

i. (1 mark)

$$\alpha + \beta = -\frac{b}{a} = -\frac{5}{2}$$

ii. (1 mark)

$$\alpha\beta = \frac{c}{a} = \frac{1}{2}$$

iii. (2 marks)

$$\begin{aligned}\alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= \frac{25}{4} - \frac{1}{2} = \frac{23}{4}\end{aligned}$$

iv. (1 mark)

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-\frac{5}{2}}{\frac{1}{2}} = -5$$

(b) (3 marks)

$$x^2 - (\alpha + \beta)x + (\alpha\beta) = 0$$

$$\alpha = 3 + \sqrt{5} \quad \beta = 3 - \sqrt{5}$$

$$\therefore x^2 - 6x + 4 = 0$$

(c) (3 marks)

$$x^4 - 17x^2 + 16 = 0$$

Let $x = 1$,

$$5 + 6 - 7 = 5 - B + 25$$

$$4 = 30 - B$$

$$\therefore B = 26$$

Hence $A = 5$, $B = 26$, $C = 25$.(e) $y = 2x^2 + kx + 8$

i. (1 mark)

$$x = 2 \quad y = 0$$

$$0 = 2(2^2) + 2k + 8$$

$$2k + 16 = 0$$

$$\therefore k = -8$$

ii. (2 marks)

$$\Delta = b^2 - 4ac > 0$$

$$k^2 - 4(2)(8) > 0 \quad k^2 - 64 > 0$$

$$(k-8)(k+8) > 0$$

$$\therefore k < -8 \quad k > 8$$

iii. (1 mark)

No, as the coefficient of x^2 is positive.**Question 3**(a) $y = (x-1)^2 - 3$

i. (1 mark)

$$V(1, -3)$$

ii. (2 marks)

$$(x-1)^2 = 4 \times \frac{1}{4}(y+3)$$

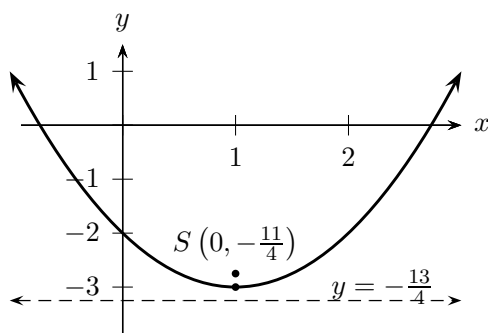
$$\therefore a = \frac{1}{4}$$

$$\therefore S\left(1, -\frac{11}{4}\right)$$

iii. (1 mark)

$$y = -3 - \frac{1}{4} = -\frac{13}{4}$$

iv. (2 marks)



(b) (2 marks)

$$P(x, y) \quad A(4, 6) \quad B(8, 10)$$

$$PA = \sqrt{(x-4)^2 + (y-6)^2}$$

$$PB = \sqrt{(x-8)^2 + (y-10)^2}$$

As $PA = PB$,

$$(x-4)^2 + (y-6)^2 = (x-8)^2 + (y-10)^2$$

$$(x-4)^2 - (x-8)^2 = (y-10)^2 + (y-6)^2$$

$$\left[(x-4) - (x-8)\right] \left[(x-4) + (x-8)\right]$$

$$= \left[(y-10) - (y-6)\right] \left[(y-10) + (y-6)\right]$$

$$4(2x-12) = -4(2y-16)$$

$$x-6 = -(y-8)$$

$$x+y = 14$$

(c) i. (2 marks)

$$P(x, y) \quad A(-3, 3) \quad B(2, -9)$$

$$PA = \sqrt{(x+3)^2 + (y-3)^2}$$

$$PB = \sqrt{(x-2)^2 + (y+9)^2}$$

$$PA = 2PB$$

$$(x+3)^2 + (y-3)^2 = 4(x-2)^2 + 4(y+9)^2$$

$$4(x-2)^2 - (x+3)^2 = (y-3)^2 - 4(y+9)^2$$

$$\left[2(x-2) - (x+3)\right] \left[2(x-2) + (x+3)\right]$$

$$= \left[(y-3) - 2(y+9)\right] \left[(y-3) + 2(y+9)\right]$$

$$(x-7)(3x-1) = (-y-21)(3y+15)$$

$$3x^2 - 22x + 7 = -3y^2 - 78y - 315$$

$$3x^2 + 3y^2 - 22x + 78y = -322$$

$$x^2 - \frac{22}{3}x + y^2 + 26y = -\frac{322}{3}$$

$$x^2 - \frac{22}{3}x + \left(\frac{11}{3}\right)^2 + y^2 + 26y + 13^2$$

$$= -\frac{322}{3} + \left(\frac{11}{3}\right)^2 + 169$$

$$\left(x - \frac{11}{3}\right)^2 + (y+13)^2 = \frac{676}{9} = \left(\frac{26}{3}\right)^2$$

ii. (2 marks)

Circle with centre $\left(\frac{11}{3}, -13\right)$ and radius $\frac{26}{3}$ **Question 4**

$$y = \frac{1}{6}(x-3)(x-1)(x+4)$$

$$= \frac{1}{6}(x^2 - 4x + 3)(x+4)$$

$$= \frac{1}{6}(x^3 - 13x + 12)$$

$$A = \int_{-4}^0 \frac{1}{6}(x^3 - 13x + 12) - (2x+2) dx$$

$$+ \int_0^4 (2x+2) - \frac{1}{6}(x^3 - 13x + 12) dx$$

$$= 2 \int_0^4 (2x+2) - \frac{1}{6}x^3 + \frac{13}{6}x - 2 dx$$

$$= 2 \int_0^4 -\frac{1}{6}x^3 + \frac{25}{6}x dx$$

$$= 2 \left[-\frac{1}{24}x^4 + \frac{25}{12}x^2 \right]_0^4$$

$$= 2 \left(-\frac{1}{24} \times 4^4 + \frac{25}{12} \times 4^2 \right) = \frac{136}{3}$$